

retenir :

dermatite irritative des langes + candidose

Diaper Dermatitis

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Objectives

- **Develop skills in observing and describing diaper dermatitis**
- **Establish a differential diagnosis**
- **Recognize diaper dermatitis associated with benign conditions**
- **Recognize diaper dermatitis associated with systemic diseases**
- **Know the treatment of the most common forms**

Diaper Dermatitis

- **One of the most common skin disorders of infancy**
- **Incidence between 7-35%, with a peak incidence at 9-12 months**

Differential diagnosis

Common Forms

- **Irritant contact dermatitis** -> la plus fréquente
- **Diaper candidiasis**
- **Seborrheic dermatitis**
- **Bullous impetigo**
- **Folliculitis**
- **Intertrigo**
- **Jacquet's dermatitis**

Differential diagnosis

Uncommon Forms

- **Psoriasis**
- **Scabies**
- **Congenital syphilis**
- **Nutritional deficiency**
- **Ecthyma gangrenosum**
- **Langerhans cell histiocytosis**
- **Granuloma gluteale infantum**
- **Chronic bullous dermatosis of Childhood**
- **Bullous mastocytosis**
- **Epidermolysis bullosa**
- **Burns**
- **Child abuse**
 - dermatite de contact allergique
 - > à différencier de la dermatite de contact irritative
 - > pas d'allergène

lésions secondaires (ex : croûte) sont une conséquence de la maladie

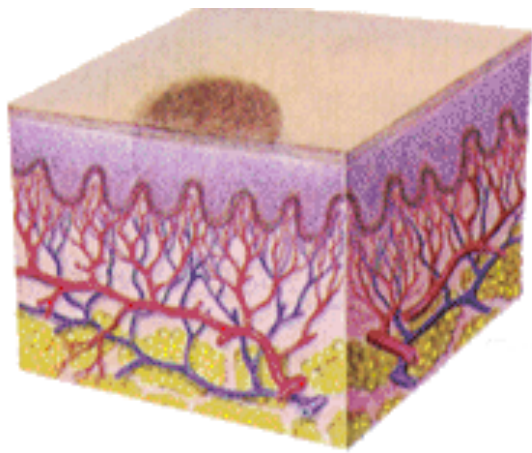
Description of skin lesions

- **Primary lesions**

- **Macule**
- **Papule**
- **Plaque**
- **Nodule**
- **Pustule**
- **Vesicle**
- **Bulla**
- **Urticaria**

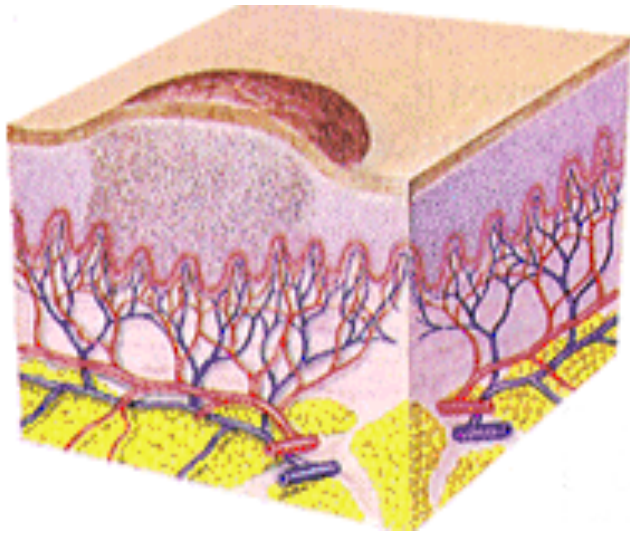
Macule

Tache cutanée circonscrite, sans relief ni infiltration, simple modification de la couleur de la peau (érythémateuse, brune, blanche, purpurique...)



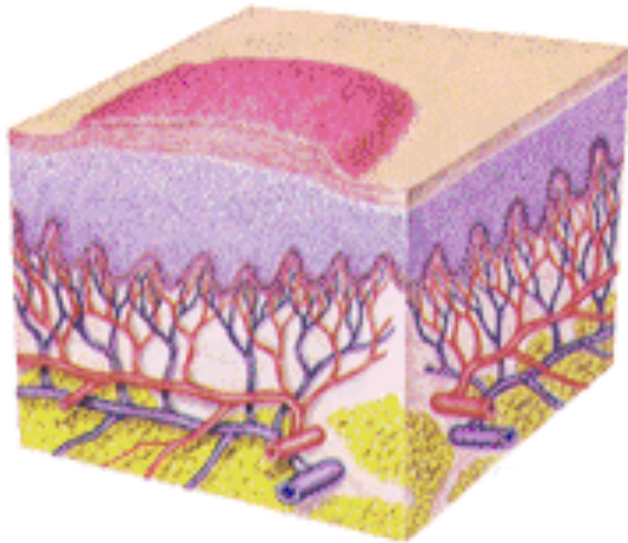
Papule

Petite lésion cutanée de <1cm de diamètre, bien délimitée, en relief, ne contenant pas de liquide



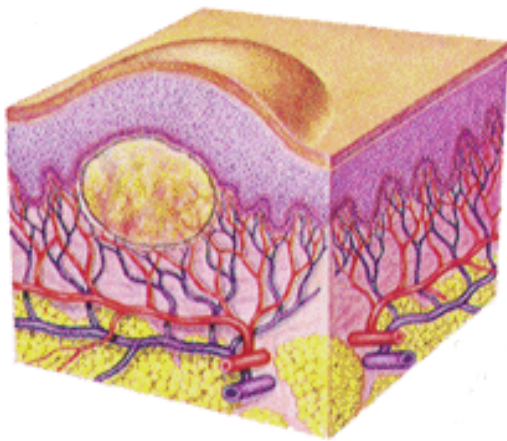
Plaque

quand lésion > 1cm de diamètre



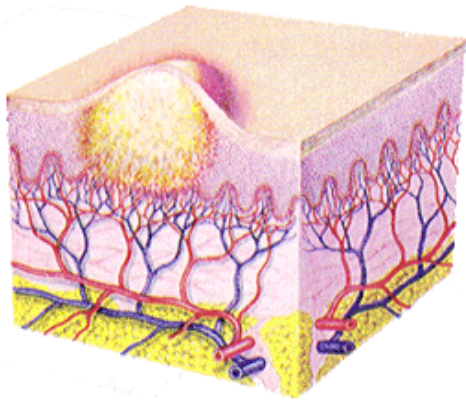
Nodule

Elevure cutanée ronde, saillante, de diamètre >1 cm, palpable, résultant soit d'un oedème, soit d'un infiltrat inflammatoire ou tumoral dermique ou hypodermique



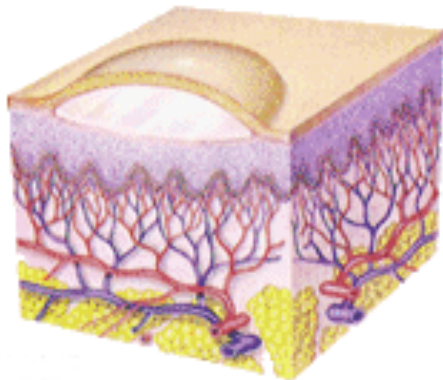
Pustule

Une **petite cavité remplie de pus** (stérile/non stérile), qui peut être folliculaire ou non folliculaire, et qui doit être distinguée d'une vésicule ou d'une bulle ne devenant purulente que secondairement



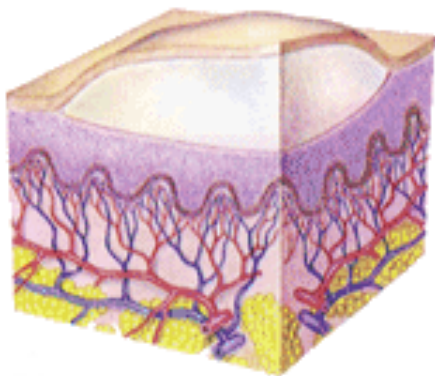
Vésicule

Une petite cavité intraépidermique à contenu liquidien, observée, de même que la spongionse, dans l'eczéma



Bulle

Soulèvement circonscrit de l'épiderme, de volume supérieur à celui d'une vésicule, qui contient un liquide clair ou séropurulent ou hémorragique



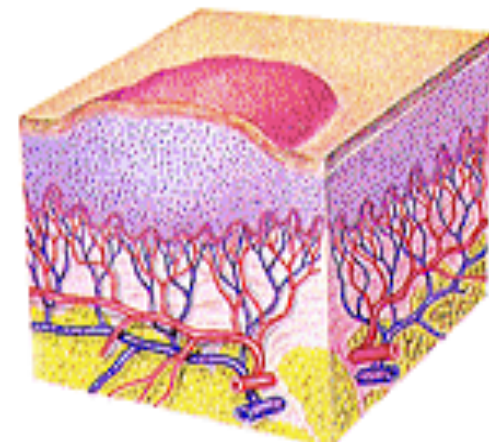
angio-oedème : lésion plus profonde qu'un urticaire

Urticaire

persiste < 24h (le plus souvent 6h)



Réaction cutanée consistant en une éruption brutale de papules prurigineuses, oedémateuses, à limites nettes, de taille, de forme et de siège très variable, dont chacune ne persiste que quelques heures



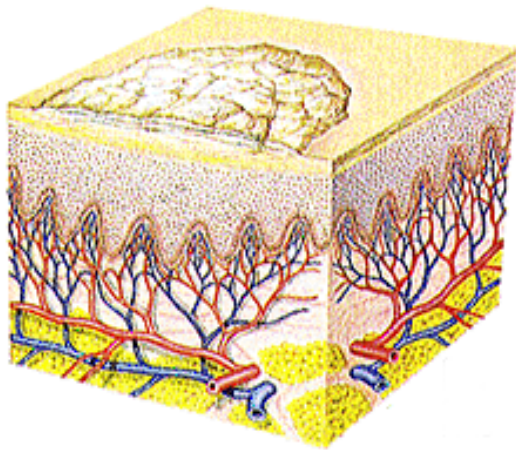
Description of skin lesions

- **Secondary lesions**

- **Squame**
- **Crust**
- **Fissure**
- **Erosion**
- **Excoriation**
- **Ulceration**
- **Atrophy**
- **Lichenification**
- **Scar**

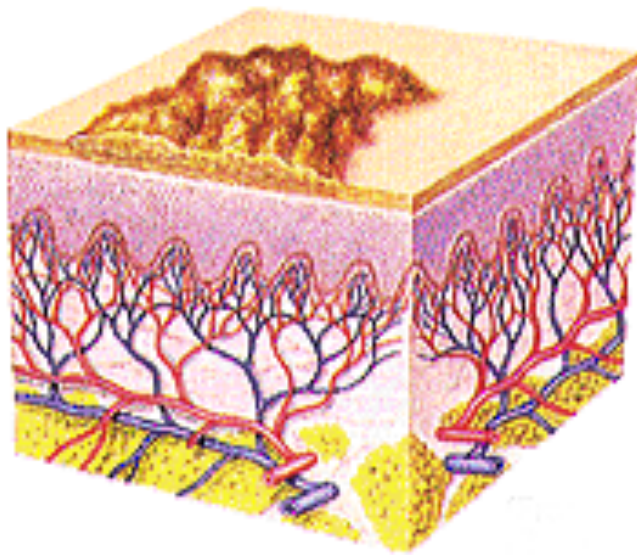
Squame = épaissement de la couche cornée

Pellicule blanchâtre qu'un grattage tangentiel doux peut détacher de la surface cutanée, et que traduit un épaissement de la couche cornée de l'épiderme qui peut être anormale (parakératosique)



Croûte

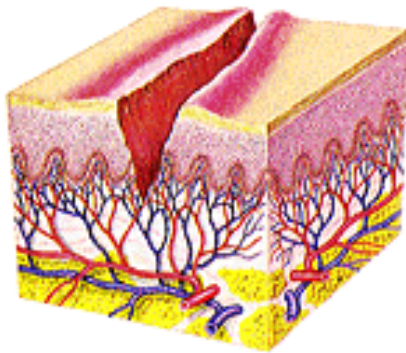
Formation dure, recouvrant une plaie ou une excoriation cutanée, facilement détachable en bloc d'un coup de curette, résultant de la dessiccation d'un exsudat séreux, hémorragique ou purulent



Rhagade (Fissure)

-> difficilement guérissable

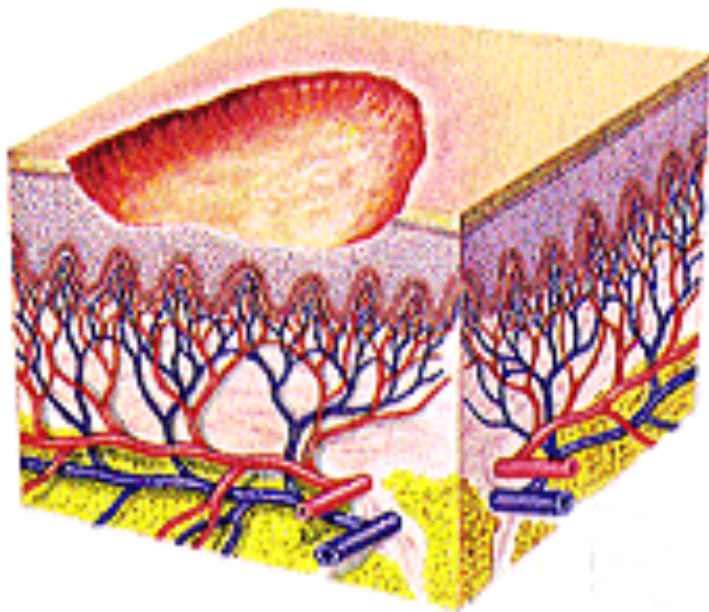
Perte de substance linéaire



de l'épiderme jusqu'à la membrane basale
cicatrice : seulement si le derme (= cuire) est touché

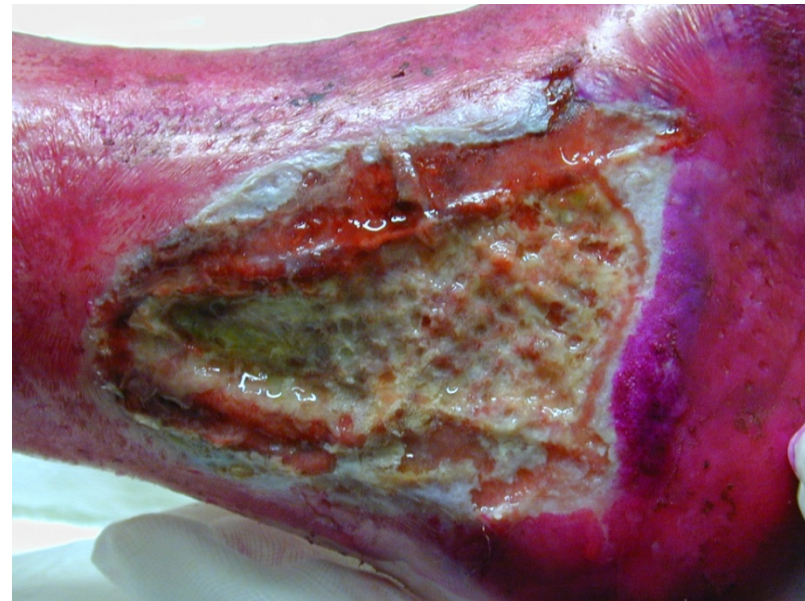
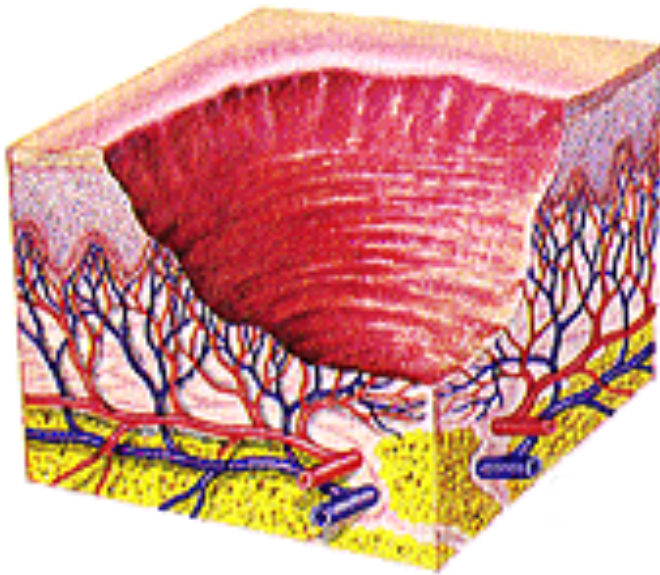
Erosion

Perte de substance cutanée ou muqueuse superficielle
n'intéressant que l'épiderme et le sommet des papilles
dermiques, et s'épidermisant sans laisser de cicatrices.



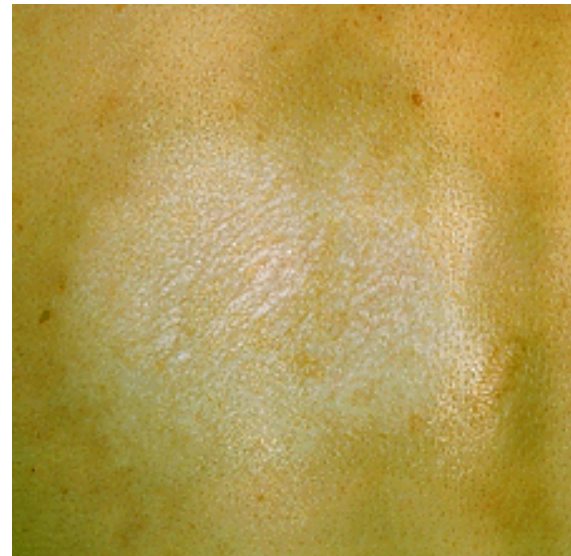
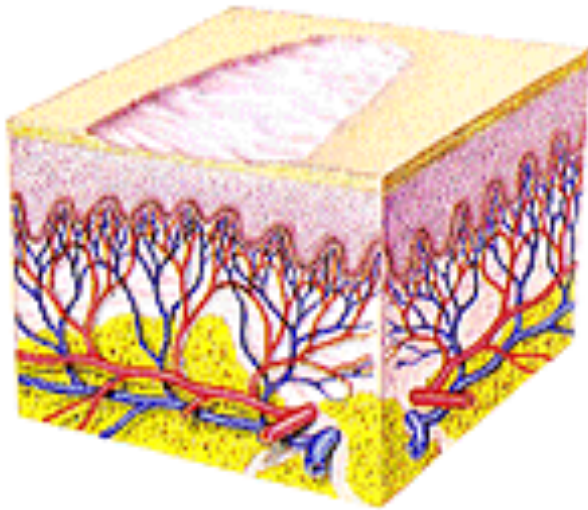
Ulcère

Perte de substance localisée résultant d'un traumatisme ou d'une nécrose, entamant au moins le derme moyen et profonde, et ne pouvant guérir sans laisser de cicatrice



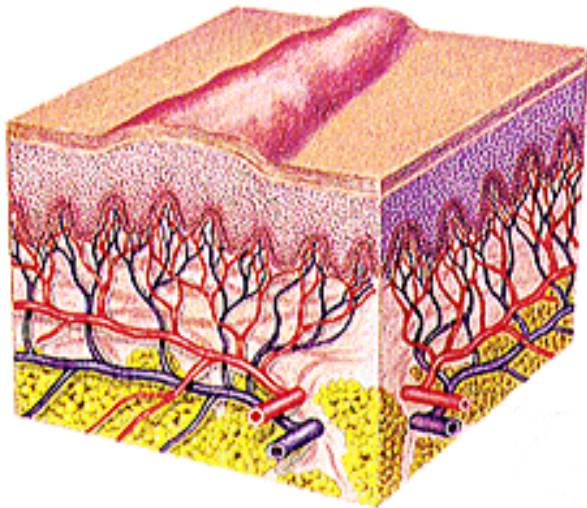
Atrophie

Modification du tégument qui est aminci, a perdu son élasticité et son relief, est devenu lisse et nacré et se ride au pincement superficiel, sauf en cas de sclérose associée



Cicatrice hypertrophique

Forme de cicatrice cutanée caractérisée par son caractère **hypertrophique**, en relief, **sans extension au-delà de la zone lésée** si c'est le cas on parle de kéloïde



lésions érythémateuse située à l'endroit d'apposition des langes (pampers)

-> peau souvent normale dans les plis inguinaux (car restent hydratés)

=> maladie irritative causée apr la friction, enzymes dans les selles, hygiène anale excessive.

Diagnosis ?



Irritant Contact Dermatitis

Cause

- **Moisture, friction, enzymes in stool** enzyme dans les selles

Treatment

ttt : application d'une pâte au zinc
-> renforce la barrière cutanée
-> haute-montagne : utilisée sur les lèvres

- **Frequent diaper changes**
- **Gently cleans** with tap water or fragrance-free diaper wipes; **avoid soaps**
- **Topical barrier cream** or ointment at all diaper changes (eg. zinc oxide ointment, petrolatum)
- **1% to 2.5% hydrocortisone** ointment or cream twice daily for active inflammation

si les autres ttt ne suffisent pas on donne alors de l'hydrocortisone
-> !! prudence car il faut être sûr du Dx
-> si candidose (2e Dx plus fréquent) + cortisone = aggravation

Diagnosis ?

-> candidose des langes

=> !! ne supporter donner de l'hydrocortisone -> aggravation



Diaper Candidiasis

Should be considered in any diaper dermatitis that develops during or shortly following antibiotic therapy

Frequently occurs in association with oral thrush

thrush = muguet = candidose

Cause

- Infection with candida species
- Primary or complicating existing irritant dermatitis
- **Diagnosis confirmed by microscopic examination** of skin scrapings or growth of yeast on Sabouraud's medium

Treatment

- Topical **antifungal** preparation: nystatin, cotrimazole
- If thrush is present: oral nystatin or fluconazole

Diaper Candidiasis

When to worry

- **Failure to respond to appropriate anti-candidal therapy warrants careful reconsideration of diagnosis**
- **Persistent or recurrent episodes of candidial infection may suggest immunodeficiency, such as chronic mucocutaneous candidiasis or human immunodeficiency virus infection**

Diagnosis ?

salmon pink patches
with greasy scales

surtout dans le
convexité et les plis
inguinaux



Seborrheic Dermatitis

- Involvement of the scalp, face, retroauricular creases, umbilicus, or chest may be present
- Begins at 3-4 weeks of age and resolves by the end of the first year of life

-> résolution spontanée dans la première, si pas le cas : 2 évolutions possibles :
-> dermatite atopique = exczema de l'enfant
-> psoriasis
-> mais on ne sait pas à l'avance dans quelle branche la maladie va évoluer

Seborrheic Dermatitis

Cause

- **Unknown**
- Associated with sebaceous gland function
- May represent an inflammatory response to the yeast *pityrosporum ovale*

Treatment

- **Skin**
 - topical low-potency corticosteroid or antifungal preparation (eg. nystatin, clotrimazole, miconazole)
- **Scalp**
 - oil massage and brushing or **anti-seborrheic shampoo**

Diagnosis ?

flaccid blister filled with clear or purulent fluid

Blister rupture rapidly



-> impetigo contagieux si Staphylocoque (réservoir narine)

Bullous impetigo

Does not usually present as a recurrent or chronic condition

Mucous membranes are not involved

Skin swab for bacterial culture

Cause

- **Infection with staphylococcus aureus that elaborates epidermolytic toxin**
- **important reservoir for staphylococci is the upper respiratory tract of asymptomatic persons**

=> il faut donc chercher et traiter les réservoirs de ces pathogènes

Bullous impetigo

Treatment

- **Topical antibiotics -> localized disease**
(fusidic acid, bacitracin, gentamicin, and erythromycin)
- **Oral anti-staphylococcal antibiotic**
(erythromycin, dicloxacillin, cephalexin, clindamycin)

Diagnosis ?



Folliculitis

Skin swab for bacterial culture

Cause

- **Infection of hair follicles with staphylococcus aureus**

Treatment

- **Gentle cleansing with antibacterial soaps**
- **Few lesions:**
 - **topical antibiotics**
(eg. fusidic acid, mupirocin)
- **Many lesions:**
 - **systemic antibiotics**
(eg. cephalexin, dicloxacillin, erythromycin)

Diagnosis ?



Intertrigo

-> peau lissée par l'humidité

-> surinfection fréquente (Candida, Strept. pyogenis)

Cause

- **Rubbing of apposed skin surfaces complicated by heat and moisture**

Treatment

- **Barrier preparation at all diaper changes to reduce moisture and friction**
- **Antifungal preparation (if candidal infection) or antibiotic (if streptococcal infection)**

Diagnosis ?



well defined shallow ulcers

Jacquet's Dermatitis

Term used to describe a severe erosive diaper eruption with ulcerated papules and nodules

Cause

- **Multiples factors including moisture, friction, enzymes on stool**
- **Considered a variant of irritant dermatitis**

Treatment

- **Topical low-potency corticosteroid combined with topical antibiotics twice daily**
- **Barrier preparation at all diaper changes**

Diagnosis ?



Perianal Streptococcal Dermatitis

Cause

- **Infection with group A beta-hemolytic streptococci**
- **Streptococcal pharyngitis may also be present**

Look for streptococcal infections in any patient presenting with new onset guttate psoriasis!

Treatment

- **Pharyngeal and skin swab for bacterial culture**
- **Oral penicillin V or erythromycin**
- **Topical antibiotics twice daily**
- **Barrier preparation at all diaper changes**

Diagnosis ?



Psoriasis

May be difficult to distinguish from seborrheic dermatitis

Presence of nail changes and red, well marginated plaques with silvery scales on the trunk, face, axillae, umbilicus, or scalp may help confirm this diagnosis

Cause

- **Unknown**
- **It probably reflects the Koebner phenomenon, triggered by trauma from exposure to stool and urine**

Psoriasis

Chronicity of psoriasis and the tendency for spontaneous remission in 38% of pediatric patients, lasting for variable time periods

5-25% will develop psoriasis within 10 years

Treatment

- **Avoid potential triggers:**
 - **injury to the skin (Koebner phenomenon), withdrawal of systemic steroids, lithium, antimalarials, beta blockers and streptococcal infections**
- **Topical emollients, low-potency corticosteroid, and calcipotriol ointment**

Diagnosis ?



Acrodermatitis enteropathica

survient souvent au moment où l'enfant passe du lait maternel au lait de vache
peu survenir plus rarement plus tôt si la mère a de la peine à transporter le zinc

May mimic a severe irritant contact dermatitis in the diaper area

Other clues

- **Unresponsive to typical therapies**
- **Infants may have sparse hair, diarrhea, or failure to gain weight**

Cause

- **Autosomal recessive disorder of zinc malabsorption**
- **Decreased serum zinc**

Acrodermatitis enteropathica

Treatment

- **Oral zinc supplementation** (zinc sulfate 5-10 mg/kg/d)
- **Topical low-potency corticosteroid**

Diagnosis ?



vesicule ou pustules

Langerhans Cell Histiocytosis

est rarissime

Areas affected

- scalp, palms, soles,
skin folds, diaper area

Other features

- hepatosplenomegaly,
lymphadenopathy
- Biopsy: S100+ and CD1a+ cells

Langerhans Cell Histiocytosis

Cause

- **Langerhans cells accumulate in the skin or other organs**

Refer to pediatric dermatologist and/or pediatric oncologist for evaluation

Treatment

- **Dictated by organ involvement**
- **Prednisone, chemotherapy, radiation, BMT**

Diagnosis ?



Ecthyma gangrenosum

-> lésion très bien délimitée

-> causée par des gram - (pseudomonas, streptocoques, ...)

Cause

- **Localized septic vasculitis usually associated with Pseudomonas aeruginosa bacteremia**
- **Diagnosis confirmed by gram stain and bacterial culture**
- **Most affected children are immunosuppressed (either congenital or acquired) or have a history of cancer chemotherapy**

Treatment

- **Systemic treatment with i.e. aminoglycoside**

Conclusions

- **Always look for clues in anamnesis and status to help elaborate the right diagnosis**
- **Prior to any consideration for therapy of diaper dermatitis, the appropriate etiology must be identified**
- **The primary goal in treating diaper dermatitis include keeping the skin dry, protected, and infection-free**
- **Refer infants if appropriate therapy fails to consider alternate diagnoses and possibly perform skin biopsy**